

LabAir 0²

In-Row Air-Conditioning Cooling System

Kabellab products are meticulously designed for easy installation while maintaining the highest quality with an aesthetically pleasing design. Kabellab combines innovative design with technology to create efficient products that achieve maximum impact. These are Kabellab's guiding principles.

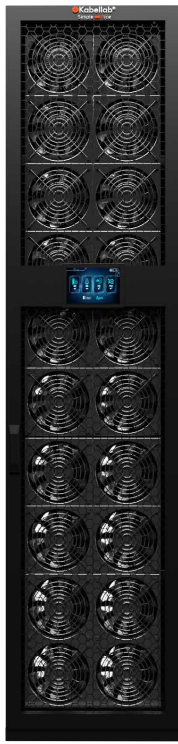


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300mm



600mm



In-Row Precision Air-Conditioner

High Efficiency and Energy Saving

- 1. High-Efficiency DC Inverter Compressor:**
Provides intelligent cooling capacity adjustment for optimal performance
- 2. Stepless Speed Fan:**
Ensures smooth and efficient airflow management.
- 3. High-Precision PID-Adjusted Water Valve (Chilled Water Type):**
Delivers precise temperature control for enhanced cooling efficiency.
- 4. Stepless Speed Scythe-Type Cooling Fan:**
Offers consistent and reliable cooling with seamless speed adjustments.
- 5. Optimal Control Policy:**
Guarantees maximum energy-saving efficiency, reducing operational costs and environmental impact.
- 6. Optimized Heat Exchanger and Air Duct Design:**
Uses Computational Fluid Dynamics (CFD) for high efficiency and low resistance in heat and mass transfer.
- 7. Ultra-High Sensible Heat Ratio:**
Prevents energy loss due to repeated humidification and dehumidification cycles.
- 8. High-Efficiency and Green Refrigerant R410A:**
Ensures environmentally friendly and efficient cooling.
- 9. High-Precision Electronic Expansion Valve:**
Provides accurate refrigerant flow control for superior system performance.



High Reliability

- 1. Uninterrupted Operation Design:**
Ensures continuous and reliable performance.
- 2. Stringent Testing and Inspection:**
All parts are rigorously tested and inspected for quality assurance.
- 3. Safe and Reliable PTC Electric Heating:**
Provides dependable and secure heating functionality..
- 4. Advanced Drain Pump:**
Features high and low water level control with double protection, including anti-liquid shock and anti-oil spill design for enhanced reliability.
- 5. Comprehensive Protection and Diagnostic Functions:**
Includes complete protection alarms and professional diagnosis capabilities.
- 6. Safety Compliance:**
Fully conforms to safety regulations for peace of mind.

Advanced Techniques

- 1. ISO Quality Management and Lean Production (TPS):**
This refers to a quality management system with 150 standards, combined with lean production principles like the Toyota Production System, ensuring efficient, high-quality IT equipment production.
- 2. Manufacturing Techniques for IT Equipment:**
This indicates specialized manufacturing processes for IT equipment, focusing on precision, durability, and compatibility with IT systems.
- 3. Fine and Decent Black Cabinet, Matching IT Room Perfectly:**
This refers to the product's sleek, professional black design, ideal for blending into standard IT environments.
- 4. High-Strength Frame, Suitable for Sea, Land, and Air Transportation:**
This feature highlights the product's durability and robust design, suitable for safe transportation by sea, land, or air, and protecting IT equipment from shipping stresses.

Intelligent Management

1. Interface design

The device features a 7" LCD touchscreen with a user-friendly interface for one-touch control, vibrant color displays of system components, and graphical trends of temperature and humidity data.

2. Protection alarm

This device features comprehensive auto protection and alarm functions, professional diagnostic capabilities, full parameter measurement and adjustment, automatic restart, water leakage detection, and lightning protection.

3. Teamwork control

Equipped with a standard R5485 communication interface and supporting the Modbus-RTU protocol for seamless integration and control of up to 32 units in a team configuration.

Convenient Maintenance

The system's architecture is meticulously crafted to ensure effortless maintenance, with full front access that simplifies any service needs. It boasts an innovative direct-connect design, integrating the fan and motor into a single unit, which not only streamlines the mechanism but also obviates the necessity for regular belt replacement—a common maintenance issue with traditional systems. Additionally, it incorporates a state-of-the-art far-infrared humidifier that operates without the need for ongoing maintenance, ensuring optimal humidity levels are maintained without additional labor or costs. This feature is particularly beneficial in environments where consistent humidity control is paramount for equipment efficiency and longevity.

Versatile Application Use

The unit boasts a versatile cooling capacity, suitable for a broad operational ambient temperature range from -40 to +55°C. It offers multiple airflow directions, various inlet and outlet piping configurations, and diverse power supply designs. The system can accommodate long pipe connections and significant height differentials. Its far-infrared humidifier is adaptable to different water conditions. The product is eco-friendly, adhering to ROHS and REACH standards, and has acquired CE, UL, and TUV certifications. It also promises quick and flexible responses to custom requirements.



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Indoor Unit

An indoor precision aircon unit maintains specific indoor climates, crucial for sensitive environments like server rooms.

1	2	3	4	5	6
HM	25	R	A	H1	P
HM	30	R	C	H1	P

No	Item	Code	Description
1	Series	HM	Link-Wind Series air conditioner for data center
2	Cooling Capacity	25	Classification
3	Type	R	In-Row
3	Cooling Mode	A	air-cooled
		W	water-cooled
		C	chilled water
		--	others
4	Air Supply	H1	front discharge
		H2	front discharge with air louver
		L1	front discharge with 90-degree left air supply
		L2	front discharge with 90-degree right air supply
		L3	front discharge with 90-degree left-right air supply
		--	others
5	Function	B	cooling
		R	cooling and heating
		H	cooling and humidification
		P	cooling, heating and humidification



Outdoor Unit

A precision aircon outdoor unit precisely regulates external temperature and humidity for sensitive equipment.

1	2	3	4	5	6	7
SW	40	N	P	H	M	0

No	Item	Code	Description
1	Series	SW	outdoor unit for data center
2	Heat Exchange	40	clarification
3	Compressor Position	N	indoor (standard)
4	Heat Exchanger Shape	P	plate (standard)
		V	V shape
5	Refrigerant	H	R410A (standard, e.g, hm25/40raw)
6	Power Supply	M	3+pe~380V 50Hz
		Q	3+pe~380V 60Hz
		--	others
7	Type	0	normal temperature (standard)
		1	low temperature



Nominal working conditions and operation range

Chilled water In-Row Precision air conditioner

Nominal Working Conditions	Air Inlet		Water Inlet and Outlet	
	dry bulb temperature	relative humidity	inlet temperature	outlet temperature
	37.8°C	20%Rh	10°C	15°C
Operation Range	indoor temperature		18~45°C	
	chilled-water inlet temperature		4~20°C	
	water side pressure		≤1.6MPa	
	altitude		≤1000, or derated	
	power adaptability		380V±10%, 50Hz±1Hz	

Note:

- Overload operation for long time can cause damage. If there is any special requirement, please specify when placing order.
- Standard power supply is 380V/50HZ/3Ph. If there is any special requirement, please specify when placing order

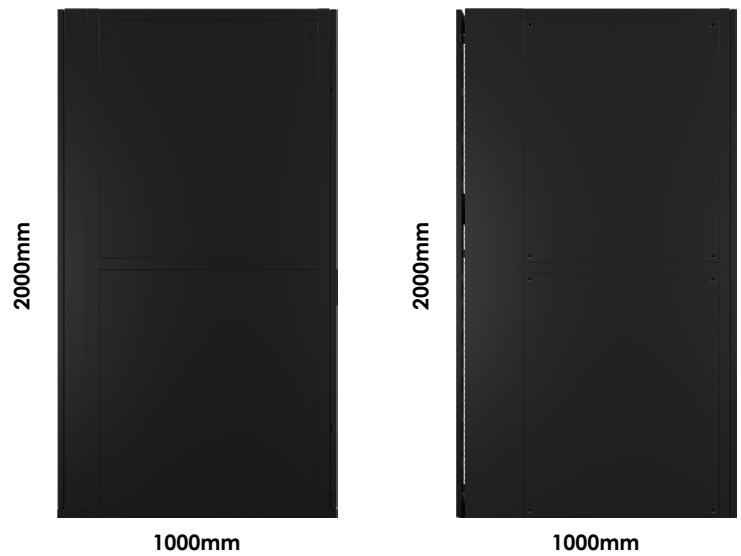
DX In-Row precision air conditioner

Nominal Working Conditions	Indoor		Outdoor		
	air inlet Status		air-cooled (air inlet Status)	water-cooled	
	dry bulb temperature	wet bulb temperature	dry bulb temperature	water inlet temperature	water outlet temperature
	37.8°C	20%RH	35°C	30°C	35°C
Operation Range	indoor temperature		18~45°C		
	outdoor temperature		-20~45°C For low temperature components, the outdoor minimum ambient temperature is 40°C		
	cooling water inlet temperature		20~34°C		
	cooling-water pressure (plate heat exchanger)		≤1.0MPa		
	altitude		≤1000, or derated		
	power adaptability		380V±10%, 50Hz±1Hz		

Note:

- Overload operation for long time can cause damage. If there is any special requirement, please specify when placing order.
- Standard power supply is 380V/50HZ/3Ph. If there is any special requirement, please specify when placing order





Specifications

Chilled water Precision air conditioner

Item	Unit	KBL-LAIR02-HM30RC	KBL-LAIR02-HM80RC
Air Flow	m ³ /h	5000	8500
Cooling Capacity	kW	30.5	61.4
Sensible Cooling Capacity	kW	30.5	61.4
Water Flow	m ³ /h	5.2	10.5
Electric heating capacity	kW	3	6
Humidifying capacity	kg/h	1.5	3
Full Load Amps (FLA)	A	29.5	15.7
Dimension	width (mm)	300	600
	depth (mm)	1000	1000
	height (mm)	2000	2000
Weight	kg	170	335

Note:

- Return air condition: dry bulb temperature 37.8°C, relative humidity 20%
- Water inlet/outlet temperature 10°C/15°C
- Due to product upgrades, specification is subject to change without prior notice.

Please confirm before ordering





Indoor Unit Specifications

DX type In-Row precision air conditioner

Item	Unit	KBL-LAIR02-HM25RA/W	KBL-LAIR02-HM40RA/W
Air Flow	m ³ /h	5000	8500
Cooling Capacity	kW	25.8	40
Sensible Cooling Capacity	kW	25.8	40
Water Flow	kW	3	6
Electric heating capacity	kg/h	1.5	3
Humidifying capacity	A	27.7	37
Dimension	width (mm)	300	600
	depth (mm)	1000	1000
	height (mm)	2000	2000
Weight	kg	240	330

Note:

- Return air condition:
dry bulb temperature 37.8°C, relative humidity 20%
- Water-cooled type working condition:
water inlet/outlet temperature 30°C/35°C
- Air-cooled type outdoor working condition:
dry bulb temperature 35°C
- Due to product upgrades, specification is subject to change without prior notice.

Please confirm before ordering





Outdoor Unit Specifications

Item	Unit	KBL-LAIR02-SW40NP	KBL-LAIR02-SW60NP
Fan Quantity	pcs	1	2
Dimension	width (mm)	1650	2010
	depth (mm)	1100	1100
	height (mm)	1150	1150
Full Load Amps (FLA)	A	2.86	5.72
Weight	kg	153	195

Note:

- Due to product upgrades, specification is subject to change without prior notice.
Please confirm before ordering

Indoor / outdoor Unit Match

Indoor Unit HM	KBL-LAIR02-HM25RA	KBL-LAIR02-HM40RA
Outdoor Unit SW	KBL-LAIR02-SW40NP	KBL-LAIR02-SW60NP
Outdoor Units Quantity	1	1

Note:

- Due to product upgrades, specification is subject to change without prior notice.
Please confirm before ordering

